

# Mass-Bound Time Scales and Interferometry

Why Holometer-Type Experiments Find No Discreteness Signal –  
a Derivation from  $\tilde{T} \cdot m = 1$

Doc. 335

27 August 2026

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## **Chapter A**

# **Mass-Bound Time Scales and Interferometry**

### Abstract

From the FFGFT fundamental relation  $\tilde{T} \cdot m = 1$  follow mass-bound Compton frequencies  $f_m = mc^2/h$  as the only physically grounded eigenfrequencies of a discrete substrate. For all stable particles these lie at least 13 orders of magnitude above the measurement frequencies of current interferometers (MHz range). This yields a precise, derivable statement about interferometry experiments for the detection of space quantisation: not a design-induced blind spot but the mass-bound nature of the time scale  $\tilde{T} = 1/m$  explains why Holometer-type experiments at their current sensitivity cannot find a discreteness signal. A  $K_{\text{frak}}$  correction to the Compton frequency of  $\Delta f_e/f_e = 100\xi_0 \approx 1.3\%$  is in principle spectroscopically accessible, but not interferometrically.

## A.1 Background: Two Approaches to Space Quantisation

Experiments such as the Fermilab E-990 Holometer ( $L = 40$  m,  $f_c = c/(2L) \approx 3.75$  MHz) and the planned QUEST experiment ( $L = 1.84$  m,  $f_c \approx 81.5$  MHz) search for a quantisation signal of spacetime. The null results of these experiments have been interpreted from various quarters as evidence for or against specific discrete substrate models.

One concrete claim is that the interferometers are by design blind at exactly the frequency at which a discrete D4 lattice with lattice constant  $a = \ell_P$  would resonate. This claim presupposes that the relevant substrate frequency is  $f \sim c/(2L)$ .

The FFGFT framework provides a more precise answer to the question of the relevant frequency scales — from first principles, without parameter fitting.

## A.2 FFGFT Eigenfrequencies from $\tilde{T} \cdot m = 1$

### The Mass-Bound Time Scale

In FFGFT, time is not an independent external quantity but is bound to mass by the fundamental relation  $\tilde{T} \cdot m = 1$  (in natural units; in SI:  $\tilde{T} \cdot m = \hbar/c^2$ ). Every massive particle therefore carries an intrinsic time scale:

$$\tilde{T}_m = \frac{\hbar}{mc^2}, \quad f_m = \frac{mc^2}{h}. \quad (\text{A.1})$$

This is the Compton frequency — not a postulate but a geometric consequence of the  $T^4$  structure and the duality relation.

### Numerical Values

| Compton frequencies of stable particles (SI) |          |                            |                           |
|----------------------------------------------|----------|----------------------------|---------------------------|
|                                              | Particle | $f_m = mc^2/h$             | $R_m = \hbar/(mc^2)$      |
|                                              | Electron | $1.2356 \times 10^{20}$ Hz | $1.288 \times 10^{-21}$ m |
|                                              | Muon     | $2.5548 \times 10^{22}$ Hz | $6.230 \times 10^{-24}$ m |
|                                              | Proton   | $2.2687 \times 10^{23}$ Hz | $7.015 \times 10^{-25}$ m |
|                                              | Neutron  | $2.2719 \times 10^{23}$ Hz | $7.006 \times 10^{-25}$ m |

### Comparison with Interferometer Frequencies

$$\frac{f_e}{f_c^{\text{Holometer}}} = \frac{1.2356 \times 10^{20} \text{ Hz}}{3.747 \times 10^6 \text{ Hz}} \approx 3.3 \times 10^{13}, \quad (\text{A.2})$$

$$\frac{f_e}{f_c^{\text{QUEST}}} \approx 1.5 \times 10^{12}. \quad (\text{A.3})$$

The smallest mass-bound frequency (electron) thus lies 13 orders of magnitude above the measurement frequencies of current interferometers.

### A.3 Lattice Dispersion Correction at the FFGFT Scale

For a discrete substrate with lattice constant  $a$ , the lattice dispersion correction to linear dispersion is of order  $(k \cdot a)^2$ . At  $k_1 = \pi/L$  this gives:

$$\left. \frac{\delta\omega}{\omega} \right|_{\text{lattice}} \sim \left( \frac{\pi a}{L} \right)^2. \quad (\text{A.4})$$

| Lattice scale $a$                  | Holometer ( $L = 40$ m) | QUEST ( $L = 1.84$ m) |
|------------------------------------|-------------------------|-----------------------|
| $\ell_P = 1.62 \times 10^{-35}$ m  | $\sim 10^{-72}$         | $\sim 10^{-70}$       |
| $R_{m_e} = 1.29 \times 10^{-21}$ m | $\sim 10^{-44}$         | $\sim 10^{-42}$       |
| $R_{m_p} = 7.0 \times 10^{-25}$ m  | $\sim 10^{-51}$         | $\sim 10^{-49}$       |

Even at the Compton scale of the electron — which in the FFGFT framework is the *physical* fundamental scale, not  $\ell_P$  — the lattice dispersion correction is  $\sim 10^{-44}$ , far beyond any measurable sensitivity.

### A.4 The $K_{\text{frak}}$ Correction and Its Measurability

The fractal correction  $K_{\text{frak}} = 1 - 100\xi_0$  with  $\xi_0 = 4/30000$  acts exclusively on absolute SI values, not on the pullback isometry itself (Doc. 190, R96, R98). Applied to the Compton frequency of the electron:

$$\Delta f_e = f_e \cdot 100\xi_0 = f_e \cdot \frac{400}{30000} \approx 1.65 \times 10^{18} \text{ Hz}. \quad (\text{A.5})$$

This corresponds to a relative deviation of  $100\xi_0 \approx 1.33\%$  from the Compton frequency. This correction is in principle spectroscopically accessible (precision spectroscopy at  $\sim 10^{20}$  Hz, i.e. hard X-ray to gamma range), but not with a Michelson interferometer in the MHz range.

### A.5 The Central Statement on Interferometry Experiments

#### Conclusion: Why Holometer-type experiments find no signal

In FFGFT the relevant frequency scale for space quantisation is not  $f \sim c/(2\ell_P)$  (universal Planck frequency) but  $f_m = mc^2/h$  (mass-bound Compton frequency). This scale differs for every particle and lies for all stable particles at least 13 orders of magnitude above the measurement frequencies of current interferometers.

It follows directly: Holometer-type experiments find nothing because the time scale of discreteness is not universal but mass-bound — not because a design-induced blind spot of the instrument lies exactly at the substrate resonance frequency.

The statement is falsifiable: an interferometer signal at frequency  $f$  would require a massless or ultralight field with  $m = hf/c^2$ . For  $f = 3.75$  MHz this gives  $m \approx 2.8 \times 10^{-44}$  kg  $\approx 3 \times 10^{-14} m_e$  — no known particle.

### A.6 Relation to the D4 Lattice Argument

The D4 lattice with lattice constant  $a = \ell_P$  has 24 nearest neighbours (8 of type  $(\pm 1, 0, 0, 0)$  and 16 of type  $(\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2})$ ). The dispersion relation of a scalar field on this lattice is:

$$\omega^2(\mathbf{k}) = \frac{c^2}{\ell_P^2} \sum_{\delta_0} (2 - 2 \cos(\mathbf{k} \cdot \delta_0 \ell_P)). \quad (\text{A.6})$$

For  $k_1 = \pi/L$  with  $L = 40$  m one has  $k_1 \ell_P \approx 1.3 \times 10^{-36}$ , hence  $k_1 \ell_P \ll 1$ : the dispersion relation is exactly linear in this regime, identical to the continuum up to order  $(k_1 \ell_P)^2 \sim 10^{-72}$ . No special lattice resonance exists at  $f_c = c/(2L)$  — the free spectral range  $c/(2L)$  is an optical property of the interferometer, not a lattice property.

FFGFT replaces the D4 lattice argument with a stronger statement: not the lattice dispersion but the mass-bound time structure  $\tilde{T} \cdot m = 1$  determines at which frequency a signal is to be expected.

*Declared 27 August 2026*